



Academic Year: 2025-26

Soundalgekar

Semester: 4

Class / Branch: SE COMPS A

Subject: OS

Name of Instructor: Prof. Shamika Mule

Name of Student: Shriya

Student ID: 24102054

Date of performance: 2/4/26

Date of submission: 2/4/26

Experiment No. 9

Aim: Write a program to implement disk scheduling algorithms FCFS, SSTF.

Program FCFS:

```
#include<stdio.h>

void main()
{
    int a[20],n,i,t=0;

    printf("Enter head pointer position:");

    scanf("%d",&a[0]);

    printf("\nEnter number of processes:");

    scanf("%d",&n);

    printf("\nEnter processes in request order");

    for(i=1;i<=n;i++)
    {
        scanf("%d",&a[i]);
    }

    for(i=0;i<n;i++)
    {
```



```
if(a[i]<a[i+1])
t+=(a[i+1]-a[i]);
else
t+=(a[i]-a[i+1]);
}
printf("\nProcessing order:");
for(i=0;i<=n;i++)
printf("\t%d",a[i]);
printf("\nTotal Head Movement:%d",t);
}Output:
```

```
apsit@apsit-HP-280-G3-MT:~/Shriya$ gcc -o xyz fcfs.c
apsit@apsit-HP-280-G3-MT:~/Shriya$ ./xyz
Enter head pointer position:50

Enter number of processes:6

Enter processes in request order67
45
89
12
48
51

Processing order:      50      67      45      89      12      48      51
Total Head Movement:199apsit@apsit-HP-280-G3-MT:~/Shriya$ |
```

Program SSTF:

```
#include <stdio.h>
#include <stdlib.h>
```

```
int main()
{
    int a[20], b[20];
    int n, i, j, k = 0;
    int head;
    int total = 0;
```



```
printf("Enter number of requests: ");
scanf("%d", &n);

printf("Enter request sequence:\n");
for(i = 0; i < n; i++)
    scanf("%d", &a[i]);

printf("Enter initial head position: ");
scanf("%d", &head);

// Add head as first element
b[k++] = head;

for(i = 0; i < n; i++)
{
    int min = 9999, pos = -1;

    for(j = i; j < n; j++)
    {
        int diff = abs(head - a[j]);

        if(diff < min)
        {
            min = diff;
            pos = j;
        }
    }

    total += min;

    // swap selected request to current position
    int temp = a[i];
    a[i] = a[pos];
    a[pos] = temp;

    head = a[i];
    b[k++] = head;
}

printf("\nProcessing order:\n");
```



```
for(i = 0; i < k; i++)  
    printf("%d\t", b[i]);  
  
printf("\nTotal Head Movement = %d\n", total);  
  
return 0;  
}
```

Output:

```
Total Head Movement:199apsit@apsit-HP-280-G3-MT:~/Shriya$ gcc -o xyz sstf.c  
apsit@apsit-HP-280-G3-MT:~/Shriya$ ./xyz  
Enter number of requests: 7  
Enter request sequence:  
12  
77  
44  
50  
32  
87  
56  
Enter initial head position: 51  
  
Processing order:  
51    50    44    32    12    56    77    87  
Total Head Movement = 114  
apsit@apsit-HP-280-G3-MT:~/Shriya$ |
```

Conclusion: Disk scheduling algorithms manage the order in which disk I/O requests are processed, with a primary goal of minimizing seek time to improve overall system performance.



Exercise 1

Program:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int req[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 54;

    int total = 0;

    // ----- FCFS -----
    int cur = head;
    printf("FCFS Order:\n%d ", cur);

    for(int i = 0; i < n; i++)
    {
        total += abs(cur - req[i]);
        cur = req[i];
        printf("-> %d ", cur);
    }

    printf("\nTotal Head Movement (FCFS) = %d\n\n", total);

    // ----- SSTF -----
    int visited[8] = {0};
    total = 0;
    cur = head;

    printf("SSTF Order:\n%d ", cur);

    for(int i = 0; i < n; i++)
    {
        int min = 9999, pos = -1;

        for(int j = 0; j < n; j++)
```



```
{
    if(!visited[j])
    {
        int diff = abs(cur - req[j]);
        if(diff < min)
        {
            min = diff;
            pos = j;
        }
    }
}

visited[pos] = 1;
total += min;
cur = req[pos];

printf("-> %d ", cur);
}

printf("\nTotal Head Movement (SSTF) = %d\n", total);

return 0;
}
```

Output:

```
apsit@apsit-HP-280-G3-MT:~/Shriya$ gcc -o xyz exp9a.c
apsit@apsit-HP-280-G3-MT:~/Shriya$ ./xyz
FCFS Order:
54 -> 98 -> 183 -> 41 -> 122 -> 14 -> 124 -> 65 -> 67
Total Head Movement (FCFS) = 631

SSTF Order:
54 -> 65 -> 67 -> 41 -> 14 -> 98 -> 122 -> 124 -> 183
Total Head Movement (SSTF) = 235
apsit@apsit-HP-280-G3-MT:~/Shriya$ |
```



Exercise 2

Program:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
int main() {
    int req[] = {45, 20, 90, 10, 50, 60, 80, 25, 70};
    int n = 9;
    int head = 50;
    int total_sstf = 0, total_scan = 0;

    // ----- SSTF -----
    int visited[9] = {0};
    int cur = head;
    printf("SSTF Order:\n%d ", cur);

    for(int i = 0; i < n; i++) {
        int min = 9999, pos = -1;
        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int diff = abs(cur - req[j]);
                if(diff < min) {
                    min = diff;
                    pos = j;
                }
            }
        }
        visited[pos] = 1;
        total_sstf += min;
        cur = req[pos];
        printf("-> %d ", cur);
    }
    printf("\nTotal Head Movement (SSTF) = %d\n\n", total_sstf);

    // ----- SCAN (towards 100 first) -----
    int cur_scan = head;
    int total = 0;
    int left[9], right[9];
```



```
int l = 0, r = 0;

for(int i = 0; i < n; i++) {
    if(req[i] < head) left[l++] = req[i];
    else right[r++] = req[i];
}

// Sort left descending
for(int i = 0; i < l-1; i++)
    for(int j = i+1; j < l; j++)
        if(left[i] < left[j]) { int t=left[i]; left[i]=left[j]; left[j]=t; }

// Sort right ascending
for(int i = 0; i < r-1; i++)
    for(int j = i+1; j < r; j++)
        if(right[i] > right[j]) { int t=right[i]; right[i]=right[j]; right[j]=t; }

printf("SCAN Order:\n%d ", cur_scan);

// Move towards 100 first
for(int i = 0; i < r; i++) {
    total += abs(cur_scan - right[i]);
    cur_scan = right[i];
    printf("-> %d ", cur_scan);
}

// Then reverse towards 0
for(int i = 0; i < l; i++) {
    total += abs(cur_scan - left[i]);
    cur_scan = left[i];
    printf("-> %d ", cur_scan);
}

printf("\nTotal Head Movement (SCAN) = %d\n\n", total_scan = total);

printf("Additional distance (SSTF - SCAN) = %d\n", total_sstf - total_scan);

return 0;
}
```




Output:

```
apsit@apsit-HP-280-G3-MT:~/Shriya$ gcc -o xyz exp9b.c
apsit@apsit-HP-280-G3-MT:~/Shriya$ ./xyz
SSTF Order:
50 -> 50 -> 45 -> 60 -> 70 -> 80 -> 90 -> 25 -> 20 -> 10
Total Head Movement (SSTF) = 130

SCAN Order:
50 -> 50 -> 60 -> 70 -> 80 -> 90 -> 45 -> 25 -> 20 -> 10
Total Head Movement (SCAN) = 120

Additional distance (SSTF - SCAN) = 10
apsit@apsit-HP-280-G3-MT:~/Shriya$ |
```